Sahaj Bhadja

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Summary

Customer-facing Software Engineer with 2+ years in **Python**, **React**, and Next.js, building REST API integrations, reproducing and fixing customer-reported issues, and shipping AI and NLP products on cloud infrastructure with CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Web Technologies, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Engineer *Los Angeles, CA*

- Built and maintained **custom integrations** for a deployed AI education platform across FastAPI services, SQL Server, PostgreSQL, and cloud storage, sustaining 99%+ reliability over 5K+ monthly jobs.
- Restructured fault-tolerant LLM orchestration with background workers and failure isolation, processing 1K+ AI jobs at 3x throughput, and cut dashboard latency from 1.2s to 350ms for 500+ users.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a **React** and Python Flask product that evaluates student writing with GPT-4o, Claude, and Qwen-VL, serving 150+ instructors and 10K+ daily requests on AWS through GitHub Actions CI/CD.
- **Reproduced** instructor-reported OCR timeouts and model failures, then shipped fallback routing and batch processing fixes, holding 90%+ grading consistency and cutting instructor feedback turnaround 60% across 10K+ handwritten and text submissions.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Worked directly with Microsoft and GitHub stakeholders in Agile sprints, taking **requests from intake through deployment** and troubleshooting large-scale API integrations to improve response times 30% for low-bandwidth users.
- Wrote 20+ automated unit and integration test suites run nightly in Jenkins across Windows, Linux, and macOS, with CSV reports and failure logs that helped the team spot recurring issues.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **Next.js** and React TypeScript platform with Node.js microservices for customer records, billing, and leads, syncing 2M+ records across locations through real-time WebSocket updates for every site.
- Debugged production errors from Sentry, Azure Monitor, and Kibana logs and shipped fixes through Jenkins CI/CD with automated tests, cutting deployment time 50% and resource wastage 42%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Served as the **technical point of contact** for the client, mapping 12+ integration points and presenting the end-to-end architecture for approval before building Flask APIs under 200ms.
- Integrated third-party payment providers through idempotent webhooks for refunds and disputes, cutting manual handling about 70%, and resolved mobile wallet connectivity issues on Android and iOS for mobile users.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, HTML, CSS
Frontend & Backend: React, Next.js, TailwindCSS, FastAPI, Flask, Node.js, REST APIs, Webhooks
AI & NLP: LLM APIs, NLP, Transformers, Embeddings, OCR, Prompt Engineering, PostgreSQL, PyTest
Cloud & Delivery: AWS, Azure, GCP, Docker, CI/CD, GitHub Actions, Jenkins, Vercel, Linux, Jest
Support: Debugging, Troubleshooting, Log Analysis, Sentry, Technical Documentation, Customer Communication, Git

Projects

AI Text Detection API SDK | *Python, TypeScript, REST APIs, pytest*

- Designed Python and TypeScript **client libraries** for an AI-text detection API with batching, retries, and rate-limit handling, plus copy-paste onboarding samples and **integration docs** for new customers.

LMS Detection Integration | *Next.js, Python, FastAPI, LTI, PostgreSQL*

- Designed an **LTI** integration that scans student submissions inside a learning management system, shows per-paragraph AI likelihood to instructors, and logs results for **institution**-level reporting and review.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, Clerk, Shadcn*

- Built an **NLP** document platform with LLaMA-2, transformer embeddings, and FAISS semantic search over 50K+ documents, reaching 95% query accuracy and cutting document review time 60% for 200+ users.